The Superior University

📝 Operating Systems Lab – Project

📌Job Scheduling for Warehouse Operation

Using SJF(Shortest Job First Non-Preemptive)

👥 Group Members

💡 *Project has been completed in a group of up to 4 members.*

List all team members with roll numbers:

- **Ayesha Akmal** **Su92-bssem-f23-269**
- **Zuha Bashir** **Su92-bssem-f23-219**
- **Muhammad Reyan** **Su92-bssem-f23-215**

📂 GitHub Repository

✅ **Both the Python code file and this documentation must be uploaded to a public GitHub repository.**

GitHub Repository Link:

https://github.com/Reyan-Da-G/Job-Scheduling-for-Warehouse-Operation-in-Phyton/tree/main or [Click here]

🔧 Scheduling Algorithm Implemented

✅ Tick the scheduling algorithm your group implemented:

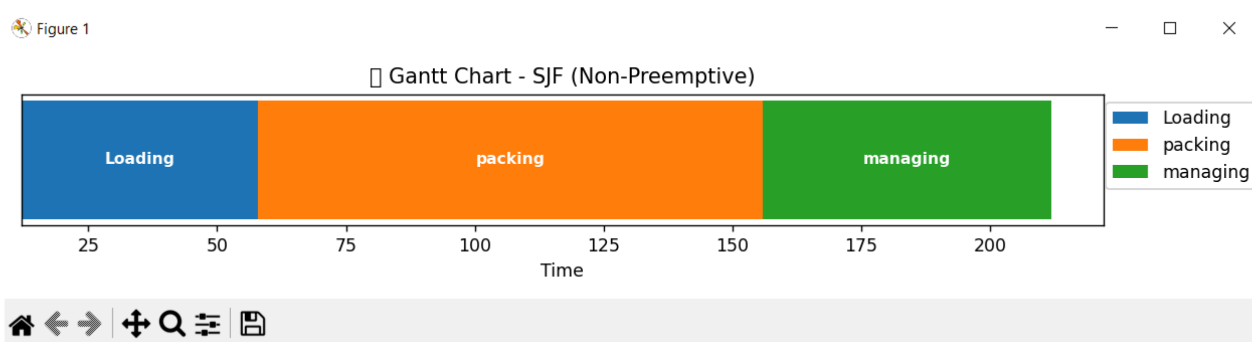
- ☑ SJF (Shortest Job First – Non-Preemptive)

📄 Project Description

This project implements job scheduling for warehouse operations using the Shortest Job First (SJF) non-preemptive algorithm. It simulates assigning jobs (like packing, loading, etc.) based on their burst time, aiming to minimize the overall waiting and turnaround time.

📸 Output Screenshots

Gantt Chart Visualization:



Output:

```
PS C:\Users\akmal\Desktop\Python>
& C:/Users/akmal/AppData/Local/Programs/Python/Python312/python.exe c:/Users/akmal/Desktop/Python/Project.py
 Enter the number of jobs: 3
 Enter job details below:

Job 1 name: Loading
   Arrival Time: 12
   Burst Time: 46

Job 2 name: packing
   Arrival Time: 45
   Burst Time: 98

Job 3 name: managing
   Arrival Time: 67
   Burst Time: 56

 Scheduled Jobs Table:

+----------+----------------+--------------+--------------+-------------------+----------------+-------------------+
| Job      |   Arrival Time |   Burst Time |   Start Time |   Completion Time |   Waiting Time |   Turnaround Time |
+----------+----------------+--------------+--------------+-------------------+----------------+-------------------+
| Loading  |             12 |           46 |           12 |                58 |              0 |                46 |
| packing  |             45 |           98 |           58 |               156 |             13 |               111 |
| managing |             67 |           56 |          156 |               212 |             89 |               145 |
+----------+----------------+--------------+--------------+-------------------+----------------+-------------------+

 Performance Metrics:
Average Waiting Time    : 34.00 units
Average Turnaround Time: 100.67 units
c:\Users\akmal\Desktop\Python\Project.py:92: UserWarning: Glyph 128338 (\N{CLOCK FACE THREE OCLOCK}) missing from font(s) DejaVu Sans.
  plt.tight_layout()
C:\Users\akmal\AppData\Local\Programs\Python\Python312\Lib\tkinter\__init__.py:862: UserWarning: Glyph 128338 (\N{CLOCK FACE THREE OCLOCK}) missing from font(s)
 DejaVu Sans.
  func(*args)
PS C:\Users\akmal\Desktop\Python>
```

🧠 Code Structure & Explanation

- **Core function:** `sjf_non_preemptive()`
- **Data structures:** list of dictionaries for jobs
- **Sorting** by arrival & burst time
- **Iterative job** selection & simulation
- **libraries:** `tabulate` or `matplotlib` (for formatting)

📊 Performance Metrics

Metric	Value
Average Waiting Time	Fill after running
Average Turnaround Time	Fill after running
Time Quantum (if RR)	N/A

🛠️ Challenges Faced

1. Sorting jobs by both arrival and burst time to respect real-time arrival.
2. Managing job selection logic when multiple jobs are available at the same time.
3. Avoiding idle time when no jobs have arrived yet.